

4.3 dplyr: Groups and Summaries

`group_by()` marks groups. `summarize()` collapses them. Always keep `n()`.

1. `summarize()` collapses rows

After `summarize()`, you keep only the summaries. Always think about `NA`. `mean(x, na.rm = TRUE)` drops missing values **inside that mean**. Filtering both delay columns first drops any row that is missing either one, so the two means use the same flights and `n()` is smaller.

```
flights |>
  summarize(mean_del = mean(dep_delay, na.rm = TRUE), n = n())
```

`n()` counts rows in the current table (or group).

2. `group_by()` makes virtual groups

`group_by(origin)` does not split the file. It marks the tibble so later verbs run **per group**. `ungroup()` clears the mark.

One grouping variable: one group per distinct value. Two variables: one group per combination.

```
flights |>
  group_by(origin) |>
  summarize(
    mean_del = mean(dep_delay, na.rm = TRUE),
    sd_del = sd(dep_delay, na.rm = TRUE),
    n = n()
  )
```

Always put `n()` in a grouped summary so you can see whether a group is tiny.

`summarize()` peels off one grouping level by default and prints a message. `.groups = "drop"` drops all of them. `options(dplyr.summarise.inform = FALSE)` quiets the message.

`arrange()` ignores groups unless you set `.by_group = TRUE`.

Pipe the summary into `ggplot2`:

```
flights |>
  group_by(origin, hour) |>
  summarize(mean_del = mean(dep_delay, na.rm = TRUE), .groups = "drop") |>
  ggplot(aes(x = hour, y = mean_del, color = origin)) +
  geom_line() +
  theme_bw()
```

3. This week, in one place

- `filter()` / `slice()` / `arrange()` for rows
- `select()` / `rename()` / `mutate()` / `relocate()` for columns
- `group_by()` then `summarize()` for one number (or a few) per group

`filter()` is the dplyr cousin of `subset()` for rows. `select()` is the cousin for columns. Note 2.1.

Chapter 5 of [R for Data Science](#) and the [data-transformation cheat sheet](#).

4. Practice

1. Standard deviation of `dep_delay`.
2. Maximum `dep_delay` at JFK, EWR, and LGA. Then do all three airports in one `group_by(origin)` pipeline.
3. For each day, the number and proportion of canceled flights, where canceled means `is.na(dep_delay) | is.na(arr_delay)`. Plot the proportion against day, and against average departure delay.